Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Applied AI engineer with **7+ years in real estate and housing SaaS** - the exact domain HomeSmart operates in. At Entrata (100K+ MAU multifamily platform), built Python data pipelines, AI-enabled workflows, and REST API integrations end to end, working directly with internal teams and external stakeholders to take solutions from prototype to production. Now at Intuit, **owning AI solution delivery**: identifying business problems with customers, choosing the right approach (AI vs. deterministic logic), building evaluation harnesses, and monitoring production quality. Built **Leasel** - an 8-agent LangGraph system for housing lease compliance with RAG, document understanding (OCR), human-in-the-loop escalation, and LLM-as-judge evaluation (93.6% precision). Proficient with **LangGraph, LangChain, Python, SQL, APIs, AWS, Docker**, and AI development tools including Claude Code. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / SQL / REST APIs | LangGraph / LangChain / Orchestration | AI Agents & RAG Pipelines | Document Understanding & OCR | Real Estate & Housing SaaS Domain | Prototype to Production Ownership | LLM Evaluation & Production Quality | Human-in-the-Loop Workflows | AWS (Lambda, S3, RDS, SQS) | Docker / Terraform / GitHub Actions CI/CD | Security & Sensitive Data Handling (PHI/HIPAA) | Data Pipelines & Large-Scale Data Systems | Claude Code / AI Development Tools | Cross-functional Communication

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - AI Solutions - Virginia, USA

- **Identify business problems and build AI solutions end-to-end:** work with stakeholders to understand workflows, define success measures, and choose the right approach - AI/ML, deterministic logic, or conventional software - based on accuracy, latency, cost, and control requirements.
- Build and deploy **production-ready AI services** in Python/FastAPI on AWS: LLM integrations, retrieval pipelines, structured-output APIs, and guardrails for sensitive financial data handling.
- Build **evaluation harnesses and regression tests** (LLM-as-judge + golden datasets) to validate AI outputs, catch hallucinations, and monitor production quality; ship fixes before they reach customers.
- Document integration patterns and build reusable components; manage **concurrent workstreams** with competing go-live deadlines, keeping technical and non-technical partners aligned.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users

- Built and owned **Python/SQL data pipelines and API integrations** end to end: from scoping with business stakeholders, through development and testing, to production monitoring and post-launch improvement.
- Delivered **large-scale data systems** processing 1.6M+ records with validation, reconciliation, and rollback checks; maintained yearly federal data update cycles with zero-downtime deployments.
- Owned **PHI/regulated document workflows** and security controls: AWS S3 migration with Libsodium encryption, HIPAA-compliant access controls, and sensitive data handling throughout the stack.
- Built Python/Bash **workflow automation** eliminating 80% of manual operational work; optimized 200+ SQL queries (30% performance gain); designed Docker microservices architecture on AWS with Terraform.
- Managed 20+ concurrent feature workstreams across 120+ Agile sprints; reviewed 300+ pull requests; worked directly with business teams and external stakeholders to translate requirements into shipped solutions.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - Pune, India

- Built housing product modules (Python, PostgreSQL) for affordable housing and rental management; delivered large-scale data import and reconciliation workflows (1.6M records).

PROJECTS

Leasel - Multi-Agent AI System for Housing Lease Compliance

Python · LangGraph · FastAPI · RAG · OCR · Human-in-the-Loop

Independently scoped and shipped an AI application for **housing lease compliance** - identifying the business problem, choosing the architecture, and taking it from prototype to a tested production system. 8-agent LangGraph orchestration: triage, document understanding (OCR ingestion), hybrid RAG retrieval, rule evaluation, confidence scoring, and **human review escalation** for low-confidence decisions. LLM-as-judge evaluation harness: 93.6% precision and 87.9% F1 across 1,240 test cases. Security controls route eligibility decisions away from the LLM entirely.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - Open-Source LLM Evaluation Toolkit

Python · FastAPI · 5 LLM Providers · REST APIs · CI/CD

15-endpoint FastAPI service integrating five AI provider APIs (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Evaluates LLM and RAG outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance using LLM-as-judge. CI regression tests with golden datasets. Demonstrates practical AI **evaluation and quality monitoring** in a production-style setup.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 I GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017